Speech Processing, Acoustic Features, Self-Supervised Learning and Machine Learning for Parkinson's Disease Analysis

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Contents

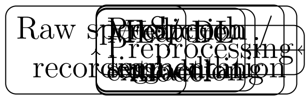
1 Introduction

Speech is a highly informative biological signal. It contains information about the linguistic content being spoken as well as information about the speaker's physiological and neurological state. Changes in the vocal folds, respiratory system, articulatory movements, motor control, and timing of speech can alter measurable properties of the acoustic waveform.

Parkinson's disease (PD) is a neurodegenerative disorder associated with motor impairments. Although the most visible symptoms involve movement, speech can also be affected. Typical speech-related changes may include reduced loudness, monopitch, reduced vocal intensity variation, imprecise articulation, altered speech timing, increased pauses, and changes in vocal-fold control.

This makes speech an attractive non-invasive signal for studying Parkinson's disease. Instead of relying only on a clinical examination, a machine-learning system can analyze recordings and estimate whether acoustic patterns are consistent with healthy speech or speech affected by PD.

The complete pipeline considered in these notes can be summarized as



The main topics covered in this report are:

- audio representation and sampling;
- speech preprocessing;
- silence and noise removal;
- voice activity detection;
- speaker normalization;
- classical acoustic features;
- pitch, jitter, shimmer, HNR and formants;
- RMS energy and loudness;
- zero-crossing rate;
- spectral analysis, FFT, DCT and cepstral representations;
- MFCCs;
- traditional machine-learning methods;
- Random Forest and Support Vector Machines;
- self-supervised learning;
- wav2vec-style speech representations;

- embeddings and deep learning;
- LSTM, Transformer and related sequence models;
- evaluation and confusion-matrix metrics.

2 Why Speech Can Be Used for Parkinson's Disease Analysis

Speech production requires coordinated control of respiration, phonation, articulation and timing. Parkinson's disease can affect the motor control needed for these processes.

Important observable effects can include:

1. **Reduced vocal loudness:** patients may speak with lower vocal intensity.
2. **Monopitch:** pitch may show less variation across an utterance.
3. **Reduced articulatory movement:** smaller movements of the lips, jaw and tongue can affect formants and spectral characteristics.
4. **Timing abnormalities:** pauses, repetitions and changes in speech rate may occur.
5. **Vocal instability:** irregular vocal-fold vibration can affect jitter, shimmer and harmonic structure.

These effects motivate extracting measurable acoustic features rather than trying to analyze the raw waveform alone.

3 Speech as an Audio Signal

An analog speech signal can be represented as a continuous-time function

$$x(t).$$

A microphone converts acoustic pressure variations into an electrical signal, which is subsequently sampled by an audio interface or recording device.

After sampling, the signal becomes a discrete sequence

$$x[n] = x(nT_s),$$

where T_s is the sampling period. The sampling frequency is

$$f_s = \frac{1}{T_s}.$$

For example, if

$$f_s = 8000 \text{ Hz},$$

then 8000 samples are stored for every second of audio.

For a recording of duration T seconds, the approximate number of samples is

$$N = f_s T.$$

Thus, a 1-second recording sampled at 16 kHz contains 16,000 samples.

3.1 Why Sampling Rate Matters

The sampling frequency determines the highest frequency that can be represented without aliasing. According to the Nyquist sampling theorem,

$$f_{\max} < \frac{f_s}{2}.$$

Therefore:

- 8 kHz sampling permits frequencies up to approximately 4 kHz;
- 16 kHz sampling permits frequencies up to approximately 8 kHz;
- 44.1 kHz sampling permits frequencies up to approximately 22.05 kHz.

Speech systems frequently use 16 kHz because it captures the frequency range needed by many speech-processing applications while keeping computational requirements manageable.

3.2 Bit Depth

Sampling frequency describes *when* measurements are taken, while bit depth describes *how accurately* each sample's amplitude is represented.

For B bits, the number of possible quantization levels is

$$2^B.$$

For example:

$$\text{16-bit} \Rightarrow 2^{16} = 65536$$

possible amplitude levels.

Higher bit depth generally provides greater amplitude resolution and a lower quantization-noise floor.

4 Raw Recording and Preprocessing

A raw recording may contain:

- silence;
- environmental noise;
- microphone noise;
- electrical interference;
- room reverberation;
- speech;
- non-speech events.

Therefore, feature extraction should generally not be performed blindly on the complete recording.

A useful preprocessing sequence is

$$\boxed{\text{Raw recording} \rightarrow \text{remove silence} \rightarrow \text{remove/reduce noise} \rightarrow \text{normalize} \rightarrow \text{segment} \rightarrow \text{extract features}}$$

The exact order can vary depending on the feature and dataset.

5 Silence Removal

Long silent intervals contribute many samples that contain little useful information about phonation or articulation.

Suppose a recording contains

$$\text{speech} - \text{silence} - \text{speech}.$$

If silence is retained, frame-level statistics can become biased toward low-energy regions.

Silence can be detected using short-time energy, RMS energy, zero-crossing rate, or a dedicated voice activity detector.

6 Noise Reduction

Noise can corrupt acoustic features. For example, high-frequency noise can change spectral measures, while broadband background noise can affect energy, HNR and zero-crossing rate.

A simple model is

$$x(t) = s(t) + n(t),$$

where $s(t)$ is the desired speech and $n(t)$ is noise.

The goal of denoising is to obtain an estimate

$$\hat{s}(t) \approx s(t).$$

Common approaches include:

- spectral subtraction;
- Wiener filtering;
- band-pass filtering;
- adaptive filtering;
- denoising neural networks.

The choice must be made carefully because aggressive denoising can remove speech characteristics that are themselves useful for diagnosis.

7 Speaker Normalization

Different speakers naturally have different recording amplitudes. One person may speak loudly while another speaks softly. Microphone placement can also change the measured amplitude.

Consequently, direct comparison of raw amplitude values may be misleading.

A common normalization is standardization:

$$x_{\text{norm}} = \frac{x - \mu}{\sigma},$$

where μ is the mean and σ is the standard deviation.

Amplitude normalization can make recordings more comparable while retaining relative variation within each recording.

7.1 Why Normalization Is Important

Consider two speakers:

$$A = [20, 18, 22]$$

and

$$B = [500, 450, 550].$$

The absolute values are very different even though their relative variation may be similar. Normalization reduces this scale difference.

Care must be taken to compute normalization statistics using only the appropriate training data when building a predictive system. Otherwise, information from validation or test recordings can leak into training.

8 Voice Activity Detection

Voice Activity Detection (VAD) determines which portions of an audio signal contain speech.

A simplified representation is

$$\text{VAD}(t) = \begin{cases} 1, & \text{speech present,} \\ 0, & \text{non-speech.} \end{cases}$$

VAD is useful for removing:

- silence;
- background-only regions;
- irrelevant non-speech intervals.

8.1 Energy-Based VAD

One simple method uses short-time RMS energy. Divide the signal into frames of length N:

$$E_{\text{RMS}}[m] = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1} x_m[n]^2}.$$

If the energy is below a threshold,

$$E_{\text{RMS}}[m] < \tau,$$

the frame may be classified as non-speech.

This approach is simple but can fail when background noise is strong or when speech is quiet.

8.2 Zero-Crossing Rate in VAD

The zero-crossing rate measures how frequently a waveform changes sign:

$$ZCR = \frac{1}{N-1}\sum_{n=1}^{N-1} \mathbf{1}\left[x[n]x[n-1] < 0\right].$$

Silence generally has low energy, while noisy regions can have high zero-crossing rates. A practical VAD may combine energy and ZCR rather than relying on one feature.

9 Framing and Windowing

Speech is non-stationary over long durations but is approximately stationary over short intervals. Therefore, speech is usually divided into short, overlapping frames.

For a frame of length N,

$$x_m[n] = x[n + mH],$$

where H is the hop size.

A window function $w[n]$ is then applied:

$$x_w[n] = x_m[n]w[n].$$

The Hamming window is commonly used:

$$w[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right).$$

Windowing reduces discontinuities at frame boundaries and therefore reduces spectral leakage.

9.1 Why Windows Should Not Be Too Long

A very long window can mix together multiple speech events and lose temporal detail. A very short window gives better temporal resolution but poorer frequency resolution.

Therefore, speech systems typically use short windows, often on the order of tens of milliseconds.

10 Fundamental Frequency and Pitch

The fundamental frequency F_0 represents the rate of vocal-fold vibration during voiced speech.

If the vocal folds complete N_c cycles during time T,

$$F_0 = \frac{N_c}{T}.$$

Pitch is the perceptual correlate of fundamental frequency.

Higher F_0 generally corresponds to a higher perceived pitch, while lower F_0 corresponds to a lower perceived pitch.

For a voiced signal with period T_0,

$$F_0 = \frac{1}{T_0}.$$

Pitch can be estimated using:

- autocorrelation;
- cepstral methods;
- harmonic analysis;
- specialized pitch-tracking algorithms.

10.1 Mean Pitch

Given frame-level pitch values

$$F_0^{(1)}, F_0^{(2)}, \dots, F_0^{(M)},$$

the mean pitch is

$$\bar{F}_0 = \frac{1}{M} \sum_{m=1}^{M} F_0^{(m)}.$$

10.2 Pitch Standard Deviation

Pitch variability can be measured using

$$\sigma_{F_0} = \sqrt{\frac{1}{M} \sum_{m=1}^{M} \left(F_0^{(m)} - \bar{F}_0 \right)^2}.$$

Reduced pitch variability can be associated with monopitch, which is commonly observed in hypokinetic dysarthria associated with Parkinson's disease.

It is important not to interpret a single pitch statistic as a diagnosis. Features should be considered collectively.

11 Jitter

Jitter describes cycle-to-cycle variation in fundamental period or frequency. It is therefore a measure of short-term frequency instability.

Let T_i denote the duration of the i-th vocal period. A simplified absolute jitter measure is

$$J_{\text{abs}} = \frac{1}{N-1} \sum_{i=1}^{N-1} |T_i - T_{i+1}|.$$

A relative version can be written approximately as

$$J_{\text{rel}} = \frac{\frac{1}{N-1} \sum_{i=1}^{N-1} |T_i - T_{i+1}|}{\frac{1}{N} \sum_{i=1}^{N} T_i}.$$

Jitter measures frequency-related irregularity, not amplitude-related irregularity.

12 Shimmer

Shimmer measures short-term variation in the amplitude of consecutive vocal-fold cycles.

If A_i is the amplitude of cycle i, an absolute shimmer-like quantity can be expressed as

$$S_{\text{abs}} = \frac{1}{N-1} \sum_{i=1}^{N-1} |A_i - A_{i+1}|.$$

A relative version is

$$S_{\text{rel}} = \frac{\frac{1}{N-1} \sum_{i=1}^{N-1} |A_i - A_{i+1}|}{\frac{1}{N} \sum_{i=1}^{N} A_i}.$$

Thus:

$$\boxed{\text{Jitter} \rightarrow \text{frequency/period variation}}$$

$$\boxed{\text{Shimmer} \rightarrow \text{amplitude variation}}$$

Both are measures of vocal stability and can be useful in pathological voice analysis.

13 Harmonics-to-Noise Ratio

Harmonics-to-Noise Ratio (HNR) quantifies the relative amount of periodic harmonic energy and aperiodic/noise-like energy in a voice signal.

A conceptual definition is

$$HNR_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{harmonic}}}{P_{\text{noise}}} \right).$$

A high HNR means that the signal is strongly periodic and harmonic, while a low HNR indicates greater noise-like or irregular components.

This is useful because irregular vocal-fold vibration can reduce the harmonic structure of speech.

HNR should not be interpreted simply as "good" or "bad" speech. It is an acoustic descriptor whose interpretation depends on the task and recording conditions.

14 Formants

Formants are resonant frequency regions of the vocal tract.

When the vocal folds generate an excitation signal, the vocal tract acts as a filter. Different configurations of the tongue, lips, jaw and pharyngeal structures produce different resonances.

The first few formants are commonly denoted

$$F_1, F_2, F_3, F_4, \ldots$$

For example:

- F_1 is strongly related to vowel height/openness;
- F_2 is strongly related to tongue frontness/backness;
- higher formants contain additional information about vocal-tract configuration.

Changes in articulatory movement can therefore produce measurable changes in formant patterns.

15 RMS Energy and Loudness

RMS energy measures the magnitude of a signal over a short window:

$$RMS = \sqrt{\frac{1}{N}\sum_{n=1}^{N} x[n]^2}.$$

Higher RMS generally indicates a stronger waveform amplitude.

Speech loudness can vary over time, and the variability itself may contain information. A system can therefore use:

- mean RMS;
- RMS standard deviation;
- minimum and maximum RMS;
- dynamic range;
- temporal patterns of energy.

Reduced vocal intensity or reduced variation in intensity can be relevant to Parkinsonian speech analysis.

16 Zero-Crossing Rate

The zero-crossing rate (ZCR) counts how frequently the waveform crosses zero.

For a frame with N samples:

$$ZCR = \frac{1}{N-1} \sum_{n=1}^{N-1} \mathbf{1}[x[n]x[n-1] < 0].$$

ZCR tends to be:

- relatively low for low-frequency periodic signals;
- higher for high-frequency components;
- often high for noise-like signals.

It is useful in speech/silence detection and as a simple spectral-shape descriptor.

17 Spectrum and Frequency-Domain Analysis

A waveform describes amplitude as a function of time. A spectrum describes how the signal's energy is distributed across frequency.

The Fourier transform represents a signal as a combination of sinusoidal components. For a continuous signal:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\, dt.$$

The magnitude spectrum is

$$|X(f)|.$$

The power spectrum can be represented as

$$P(f) = |X(f)|^2.$$

The spectrum therefore answers questions such as:

> Which frequencies contain most of the signal's energy?

This is especially useful for speech because different phonemes and vocal qualities produce different spectral patterns.

18 DFT and FFT

For a discrete signal with N samples, the Discrete Fourier Transform is

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}.$$

The inverse DFT is

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}.$$

The Fast Fourier Transform (FFT) is an efficient algorithm for computing the DFT.
A direct DFT requires approximately

$$O(N^2)$$

operations, while FFT algorithms can reduce the complexity to approximately

$$O(N \log N).$$

This efficiency is why FFT is routinely used in speech processing.

19 Discrete Cosine Transform

The Discrete Cosine Transform (DCT) represents a sequence using cosine basis functions.
One common DCT-II definition is

$$C_k = \sum_{n=0}^{N-1} x_n \cos\left[\frac{\pi}{N}\left(n + \frac{1}{2}\right)k\right].$$

The DCT is particularly important in MFCC computation.

The reason is that filter-bank energies are highly correlated. The DCT approximately decorrelates these values and concentrates much of the useful information into a smaller number of coefficients.

20 Cepstrum

The cepstrum is obtained by taking the inverse Fourier transform of the logarithm of the magnitude spectrum:

$$c(\tau) = \mathcal{F}^{-1}\left\{\log|X(f)|\right\}.$$

The use of the logarithm converts multiplicative spectral effects into additive components.

For example, a simplified speech production model is

$$X(f) = G(f)V(f),$$

where $G(f)$ represents excitation and $V(f)$ represents the vocal-tract filter. Taking logarithms gives

$$\log|X(f)| = \log|G(f)| + \log|V(f)|.$$

The cepstral representation can therefore help separate source-related and filter-related characteristics.

21 MFCC

Mel-Frequency Cepstral Coefficients (MFCCs) are among the most widely used traditional speech features.

The typical MFCC pipeline is

Waveform → Framing → Windowing → FFT → Power Spectrum → Mel Filter Bank → log → DCT →

21.1 Mel Scale

The mel scale is designed to approximate human auditory frequency perception.

A common conversion is

$$m(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right).$$

The inverse conversion is

$$f = 700\left(10^{m/2595} - 1\right).$$

21.2 Mel Filter Bank

The power spectrum is passed through a set of overlapping triangular filters. Each filter aggregates energy from a frequency region.

If $P[k]$ is the power spectrum and $H_m[k]$ is the m-th filter,

$$E_m = \sum_k P[k]H_m[k].$$

Then logarithmic filter-bank energies are computed:

$$L_m = \log(E_m + \epsilon).$$

Finally, the DCT is applied:

$$c_n = \sum_{m=1}^{M} L_m \cos\left[\frac{\pi n}{M}\left(m - \frac{1}{2}\right)\right].$$

The resulting c_n values are the MFCCs.

21.3 Why MFCCs Are Useful

MFCCs capture broad spectral characteristics associated with speech production. They are compact, relatively robust and computationally inexpensive.

However, MFCCs do not explicitly encode every clinically relevant property. Therefore, additional features such as pitch, jitter, shimmer, HNR, energy and formants may provide complementary information.

22 Traditional Acoustic Features

A classical feature vector can be constructed as

$$\mathbf{x} = [F_0,\, \sigma_{F_0},\, \mathit{Jitter},\, \mathit{Shimmer},\, HNR,\, RMS,\, ZCR,\, F_1,\, F_2,\, F_3,\, MFCC_1, \ldots, MFCC_K].$$

The model then receives this numerical vector rather than the complete audio waveform.

A major advantage is interpretability: each feature has a recognizable acoustic meaning.

A disadvantage is that feature engineering may fail to capture complex patterns that are not represented explicitly.

23 Classical Machine Learning

After feature extraction, traditional machine-learning models can be trained using the feature vector.

The general pipeline is

$$\text{Audio} \rightarrow \text{Acoustic features} \rightarrow \text{Feature vector} \rightarrow \text{ML classifier} \rightarrow \text{Prediction}.$$

Models mentioned in the notes include:

- Random Forest;
- Support Vector Machine;
- feed-forward neural networks.

24 Random Forest

A Random Forest is an ensemble of decision trees.

Suppose there are B trees:

$$T_1, T_2, \ldots, T_B.$$

For classification, each tree produces a class prediction. The final output can be obtained by majority voting:

$$\hat{y} = \operatorname{mode}\left\{T_1(\mathbf{x}), \ldots, T_B(\mathbf{x})\right\}.$$

Each tree is trained using randomness in:

- the training samples;
- the subset of features considered at splits.

This reduces correlation between trees and often improves generalization.

24.1 Advantages

- handles nonlinear relationships;
- relatively robust;
- can estimate feature importance;
- requires limited feature scaling.

24.2 Limitations

- can become computationally expensive with many trees;
- feature importance can be biased in some settings;
- does not automatically learn useful representations directly from raw waveforms.

25 Support Vector Machine

A Support Vector Machine (SVM) seeks a decision boundary that maximizes the margin between classes.

For a linear classifier,

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b.$$

The decision is based on the sign of $f(\mathbf{x})$.

The margin is related to

$$\frac{2}{\|\mathbf{w}\|}.$$

A soft-margin SVM solves an optimization problem of the form

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i$$

subject to

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0.$$

The parameter C controls the trade-off between margin size and training errors.

SVMs can also use kernels to represent nonlinear decision boundaries.

26 Feed-Forward Neural Network

A feed-forward neural network receives a feature vector and transforms it through several nonlinear layers.

For example,

$$\mathbf{h}_1 = \sigma(W_1\mathbf{x} + b_1),$$

$$\mathbf{h}_2 = \sigma(W_2\mathbf{h}_1 + b_2),$$

$$\hat{\mathbf{y}} = W_3\mathbf{h}_2 + b_3.$$

For binary classification, a sigmoid can produce

$$p(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-z}}.$$

The model parameters are learned by minimizing a loss such as binary cross-entropy:

$$\mathcal{L} = -\left[y \log p + (1 - y) \log(1 - p)\right].$$

27 Self-Supervised Learning

Self-supervised learning (SSL) learns representations from data without requiring manual labels for every training example.

The central idea is to create a learning signal from the data itself.

Instead of:

$$\text{audio} \rightarrow \text{human label},$$

an SSL system can learn from relationships within the audio.

This is especially valuable for speech because large collections of unlabeled speech are easier to obtain than large collections of carefully annotated clinical speech.

27.1 Why SSL Is Useful

Traditional acoustic features explicitly define what information should be measured. SSL instead attempts to learn a representation containing useful patterns automatically.

A learned representation can potentially encode:

- phonetic information;
- speaker characteristics;
- prosody;
- temporal patterns;
- acoustic regularities;
- higher-level speech structure.

For a downstream Parkinson's disease task, a pretrained speech representation can be followed by a classifier.

$$\boxed{\text{Unlabeled speech} \rightarrow \text{SSL pretraining} \rightarrow \text{speech embedding} \rightarrow \text{PD classifier}}$$

28 wav2vec-Style Representation Learning

wav2vec and related models learn useful representations directly from speech waveforms.

A simplified conceptual architecture contains:

$$\text{waveform} \rightarrow \text{feature encoder} \rightarrow \text{context network} \rightarrow \text{contextual representation}.$$

The representation can then be used by a downstream model.

One important distinction is that a pretrained representation does not necessarily correspond to a single human-interpretable feature such as "pitch" or "jitter". Instead, the embedding is distributed across many dimensions.

For example, if an encoder outputs

$$\mathbf{z} \in \mathbb{R}^{768},$$

then the 768 dimensions collectively form a learned representation.

It is therefore better to think of an embedding as a learned coordinate system for speech information rather than as 768 independent hand-designed acoustic features.

29 Hand-Crafted Features vs Learned Embeddings

The two approaches can be compared as follows.

Table 1: Comparison of traditional and learned speech representations.

Property	**Hand-crafted features**	**Learned embeddings**
Examples	MFCC, pitch, jitter, shimmer, HNR	wav2vec-style embeddings
Feature design	Human-designed	Learned from data
Interpretability	Usually high	Usually lower
Dimensionality	Often compact	Often high-dimensional
Data requirement	Lower	Usually higher for training from scratch
Complex patterns	Limited by feature design	Can capture complex patterns
Domain transfer	May require redesign	Pretraining can improve transfer

A useful practical strategy is to compare both approaches and, when justified, combine them.

30 Combining Learned and Hand-Crafted Features

Suppose a neural speech encoder produces

$$\mathbf{e} \in \mathbb{R}^{768},$$

and classical acoustic analysis produces

$$\mathbf{a} \in \mathbb{R}^K.$$

They can be concatenated:

$$\mathbf{h} = [\mathbf{e}; \mathbf{a}] \in \mathbb{R}^{768+K}.$$

A classifier can then learn from both:

$$\hat{y} = f_\theta(\mathbf{h}).$$

This can be useful because the embedding may contain broad speech information while the hand-crafted features explicitly represent clinically meaningful properties.

However, feature scaling and dimensionality must be considered carefully.

31 Sequence Models

Speech is inherently sequential. A person's current acoustic state is related to surrounding frames.

Therefore, rather than treating every frame independently, sequence models can process

$$\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T.$$

Two important families are LSTMs and Transformers.

32 LSTM

Long Short-Term Memory (LSTM) networks are recurrent neural networks designed to model temporal dependencies.

The LSTM contains gates controlling information flow.

The forget gate is

$$f_t = \sigma(W_f[x_t, h_{t-1}] + b_f).$$

The input gate is

$$i_t = \sigma(W_i[x_t, h_{t-1}] + b_i).$$

The candidate cell state is

$$\tilde{c}_t = \tanh(W_c[x_t, h_{t-1}] + b_c).$$

The cell state is updated by

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t.$$

The output gate is

$$o_t = \sigma(W_o[x_t, h_{t-1}] + b_o),$$

and the hidden state is

$$h_t = o_t \odot \tanh(c_t).$$

The cell state provides a mechanism for retaining information over multiple time steps.

33 Transformer for Speech

Transformers model sequences primarily through attention rather than recurrence.

Given query, key and value matrices Q, K, V, scaled dot-product attention is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

The model can therefore determine which time positions are relevant to each other.

For speech, this can help capture relationships between distant portions of an utterance.

33.1 Multi-Head Attention

Instead of one attention operation, a Transformer uses multiple attention heads:

$$head_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V).$$

The heads are concatenated and projected:

$$MultiHead(Q, K, V) = Concat(head_1, \ldots, head_h)W^O.$$

34 Embeddings

An embedding is a numerical vector representing an object in a learned feature space.

For speech,

$$\text{audio} \to \text{encoder} \to \mathbf{e}.$$

For example,

$$\mathbf{e} \in \mathbb{R}^{768}.$$

A downstream classifier can use this vector:

$$\mathbf{e} \to \text{MLP} \to P(\text{PD}).$$

Embeddings are powerful because the model can learn combinations of acoustic patterns that may not correspond to individual hand-crafted measurements.

35 Model Training and Loss

For supervised classification, the model predicts a probability

$$p_\theta(y|\mathbf{x}).$$

The parameters θ are learned by minimizing a loss.
For binary classification, binary cross-entropy is

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log p_i + (1-y_i)\log(1-p_i)\right].$$

The training procedure is conceptually:

$$\text{input} \rightarrow \text{model} \rightarrow \text{prediction} \rightarrow \text{loss} \rightarrow \text{backpropagation} \rightarrow \text{parameter update.}$$

For an optimizer with learning rate η,

$$\theta \leftarrow \theta - \eta\nabla_\theta\mathcal{L}.$$

36 Train, Validation and Test Sets

A predictive system should separate data into different subsets.

- **Training set:** used to learn model parameters.
- **Validation set:** used to select hyperparameters and model configurations.
- **Test set:** used for final evaluation.

For subject-based clinical data, splitting must be done carefully. If recordings from the same person appear in both training and test sets, the model may learn speaker identity rather than disease-related characteristics.

A subject-independent split is therefore generally preferable when the goal is to estimate performance on unseen patients.

37 Class Imbalance

Suppose a dataset contains

$$90 \text{ healthy samples}$$

and

$$10 \text{ PD samples.}$$

A classifier that predicts "healthy" for every sample achieves 90% accuracy, despite having no ability to detect PD.

Therefore, accuracy alone can be misleading.

Possible solutions include:

- class-weighted loss;
- oversampling;
- undersampling;
- balanced batch construction;
- appropriate evaluation metrics.

38 Confusion Matrix

For binary classification, the confusion matrix contains:

- True Positive (TP);
- True Negative (TN);
- False Positive (FP);
- False Negative (FN).

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

39 Evaluation Metrics

39.1 Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}.$$

Accuracy measures the overall fraction of correct predictions.

39.2 Precision

$$Precision = \frac{TP}{TP + FP}.$$

Precision answers:

> Among samples predicted as positive, how many were actually positive?

39.3 Recall / Sensitivity

$$Recall = \frac{TP}{TP + FN}.$$

Recall answers:

> Among the truly positive samples, how many did the model detect?

For disease screening, recall can be particularly important because false negatives may be costly.

39.4 Specificity

$$Specificity = \frac{TN}{TN + FP}.$$

Specificity measures the ability to correctly identify negative cases.

39.5 F1 Score

The F1 score is

$$F_1 = 2\frac{Precision \cdot Recall}{Precision + Recall}.$$

It balances precision and recall.

40 ROC Curve and AUC

A classifier often outputs a probability rather than an immediate class decision.

Changing the decision threshold changes:

- true positive rate (TPR);
- false positive rate (FPR).

The ROC curve plots

$$TPR = \frac{TP}{TP + FN}$$

against

$$FPR = \frac{FP}{FP + TN}.$$

The Area Under the ROC Curve (AUC) summarizes discrimination across thresholds.

An AUC close to 1 indicates strong separation, while an AUC near 0.5 is similar to random ranking.

41 Why Accuracy Alone Is Not Enough

For medical or clinical prediction, a model should not be judged solely by accuracy.

For example, if

$$TP = 9, \quad FN = 1,$$

then

$$Recall = \frac{9}{10} = 0.9.$$

But if there are many false positives, precision may be low.

Therefore, a good evaluation should report several complementary measures, such as

$$Accuracy, \quad Precision, \quad Recall, \quad Specificity, \quad F1, \quad AUC.$$

Confidence intervals and subject-independent evaluation are also valuable when the dataset is small.

42 Feature Importance and Interpretability

Traditional models can provide interpretable information about which features contribute to prediction.

For example, a Random Forest can provide feature-importance estimates. If features such as jitter, shimmer, HNR, pitch variability and MFCCs are highly informative, this can motivate further investigation.

However, feature importance should not automatically be interpreted as causal evidence.

Similarly, an embedding dimension having a strong relationship with the prediction does not mean that the dimension directly represents a particular clinical variable.

Interpretability methods such as permutation importance, SHAP, saliency methods or feature ablation can provide additional evidence.

43 A Complete Practical Pipeline

A robust end-to-end system can be organized into the following stages.

43.1 Stage 1: Data Collection

Collect speech recordings with reliable metadata and labels.

43.2 Stage 2: Quality Control

Check:

- corrupted files;
- sampling rates;
- clipping;
- recording duration;
- extreme background noise;
- duplicate recordings.

43.3 Stage 3: Preprocessing

Perform appropriate:

- resampling;
- silence removal;
- noise reduction;
- amplitude normalization.

43.4 Stage 4: VAD and Segmentation

Identify speech-containing intervals and divide them into frames or utterance segments.

43.5 Stage 5: Feature Extraction

Extract either:

$$\text{hand-crafted features}$$

or

$$\text{learned speech representations.}$$

A combined representation is also possible.

43.6 Stage 6: Model Training

Train an SVM, Random Forest, MLP, LSTM, Transformer or another suitable model.

43.7 Stage 7: Validation

Tune hyperparameters using the validation set without using the test set.

43.8 Stage 8: Testing

Evaluate once on the held-out test subjects.

43.9 Stage 9: Clinical Interpretation

Report appropriate metrics and investigate whether the model is learning generalizable speech patterns rather than recording-specific artifacts.

44 Important Conceptual Distinctions

44.1 Pitch vs Fundamental Frequency

Fundamental frequency is a physical/acoustic quantity measured in Hz. Pitch is the perceptual correlate of that frequency.

44.2 Jitter vs Shimmer

$$\boxed{\textit{Jitter} = \text{cycle-to-cycle frequency/period variation}}$$

$$\boxed{\textit{Shimmer} = \text{cycle-to-cycle amplitude variation}}$$

44.3 RMS vs Loudness

RMS is an objective signal-amplitude measure. Perceived loudness is a psychoacoustic quantity and depends on frequency content and human hearing. Therefore RMS and loudness should not be treated as mathematically identical.

44.4 FFT vs DCT

FFT computes a Fourier representation of the signal. DCT represents the signal using cosine basis functions. MFCC computation commonly uses FFT first and DCT later.

44.5 MFCC vs Embedding

MFCCs are explicitly designed acoustic features. A neural embedding is learned from data and can contain many kinds of information.

44.6 VAD vs Noise Removal

VAD determines *where speech exists*. Noise reduction attempts to *reduce unwanted acoustic components*. They are different operations.

45 Potential Sources of Error

Speech-based clinical modeling has several challenges.

1. **Speaker variability:** age, sex, accent and natural voice differences can be large.
2. **Microphone variability:** recording hardware changes the measured spectrum.
3. **Room acoustics:** reverberation can alter acoustic features.
4. **Background noise:** noise can distort jitter, shimmer, HNR and spectral measurements.
5. **Class imbalance:** an uneven dataset can make accuracy misleading.
6. **Data leakage:** recordings from the same speaker in different splits can inflate performance.
7. **Small clinical datasets:** deep models can overfit.
8. **Confounding variables:** the model may learn age, sex, microphone characteristics or recording protocol instead of PD-related information.

These issues should be considered before interpreting a high classification score as evidence of clinical usefulness.

46 Recommended Experimental Comparison

A useful experimental study can compare several feature/model combinations.

Table 2: Example experimental design.

Experiment	Representation	Classifier
1	MFCC only	SVM
2	Classical acoustic features	Random Forest
3	Classical acoustic features	MLP
4	Learned speech embedding	MLP
5	Learned speech embedding	SVM
6	Classical + learned features	MLP
7	Frame sequence	LSTM
8	Frame sequence	Transformer

This comparison helps answer an important research question:

> Does learned representation provide information beyond carefully designed traditional acoustic features?

47 Summary of the Main Ideas

The central ideas from the notes can be summarized as follows.

1. A microphone converts speech into an audio signal.
2. Sampling converts the continuous signal into discrete samples.
3. Preprocessing reduces silence, noise and recording-level variability.
4. VAD identifies speech-containing regions.
5. Framing makes speech approximately stationary over short intervals.
6. FFT provides frequency-domain information.
7. DCT is important for cepstral feature compression and decorrelation.
8. MFCCs summarize the spectral envelope using the mel scale.
9. Pitch/F0 describes vocal-fold vibration rate.
10. Jitter measures short-term frequency/period irregularity.
11. Shimmer measures short-term amplitude irregularity.
12. HNR measures harmonic energy relative to noise-like energy.
13. Formants describe resonances of the vocal tract.

14. RMS describes short-time signal magnitude.
15. ZCR measures sign changes in the waveform.
16. Classical ML models can operate on engineered acoustic features.
17. SSL models learn speech representations from large amounts of unlabeled data.
18. wav2vec-style models provide high-dimensional learned embeddings.
19. LSTMs and Transformers can model temporal speech information.
20. Evaluation should use confusion-matrix-derived metrics rather than accuracy alone.
21. Subject-independent splitting is essential for a credible Parkinson's disease experiment.

48 Conclusion

Speech processing provides a rich framework for studying Parkinson's disease because neurological and motor changes can influence vocal production and speech timing.

A traditional approach explicitly measures clinically meaningful acoustic properties such as pitch, jitter, shimmer, HNR, formants, energy and MFCCs. These features are compact and comparatively interpretable, making models such as Random Forest and SVM attractive for smaller datasets.

A modern approach uses self-supervised speech models to learn representations directly from large amounts of speech. These embeddings can capture complex patterns that are difficult to describe with a small set of hand-crafted features. Sequence models such as LSTMs and Transformers can additionally model temporal relationships across speech frames.

The strongest experimental design is not necessarily the most complicated model. A reliable pipeline requires careful preprocessing, subject-independent data splitting, avoidance of leakage, appropriate normalization, balanced evaluation and analysis of potential confounding factors.

Thus, the overall conceptual progression is:

$$\boxed{\text{Speech} \rightarrow \text{Preprocessing} \rightarrow \text{Representation} \rightarrow \text{Learning} \rightarrow \text{Evaluation}}$$

where representation may be either hand-crafted acoustic features, self-supervised embeddings, or a carefully justified combination of both.